

Types of Scientific Research

Observational Research

Descriptive Research

Describes individuals or populations

Case Studies

- Track a single person or group of people over time
- Discuss characteristics of or events that occur to a single person or group
- Primary limitation:
 - May not be generalizable to larger populations

Surveys

- Questionnaires or interviews administered to samples of people
- Primary limitations:
 - Self-reported data, and can be subject to bias
 - Potential issues with the reliability, validity, or how wording of questions can influence people's answers



Historical Research

- Looks at events that happened in the past and uses that data to make inferences about other times, including the future
- Limitations:
 - Information from the past may be incomplete or inaccessible
 - May be subject to the biases of the researchers involved



Exploratory or Epidemiological Research

Looks for relationships between variables in large populations

Cross-sectional Studies

- Track variables in a cross-section of a population at a specific point of time
- Most useful for:
 - Population-based surveys
 - To assess the prevalence of diseases in a population
- Primary limitation:
 - Impossible to assess causal relationships because it's a one-time "snapshot" measurement

Case-control Studies

- Determine if exposure to something is associated with an outcome
- Scientists identify people with the outcome and similar people who don't have the outcome (aka the control group)
- Primary limitation:
 - Uses data that has already been collected or occurred in the past

Cohort Studies

Group (cohort) followed over time to determine association between an exposure and an outcome

- Researchers:
 - Identify cohort free of outcome (e.g., disease)
 - Identify exposure or event of interest
 - Follow over time until outcome occurs
 - Assess association between exposure and risk of outcome
- Limitations:
 - Susceptibility to selection bias
 - Subjects are followed for extended periods of time and may withdraw or be lost follow-up
 - If retrospective: less control over variables, and subjects may be susceptible to recall bias